

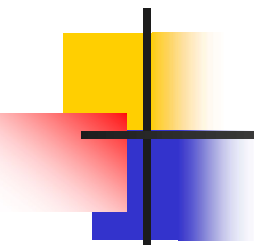
FLOW CONTROL

*The most important responsibilities of the data link layer are **flow control** and **error control**. Collectively, these functions are known as **data link control**.*

- An important design issue that occurs in the data link layer (and higher layers as well) is what to do with a sender that systematically wants to transmit frames faster than the receiver can accept them.
- This situation can occur when the sender is running on a fast, powerful computer and the receiver is running on a slow, low-end machine.

Note

Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.



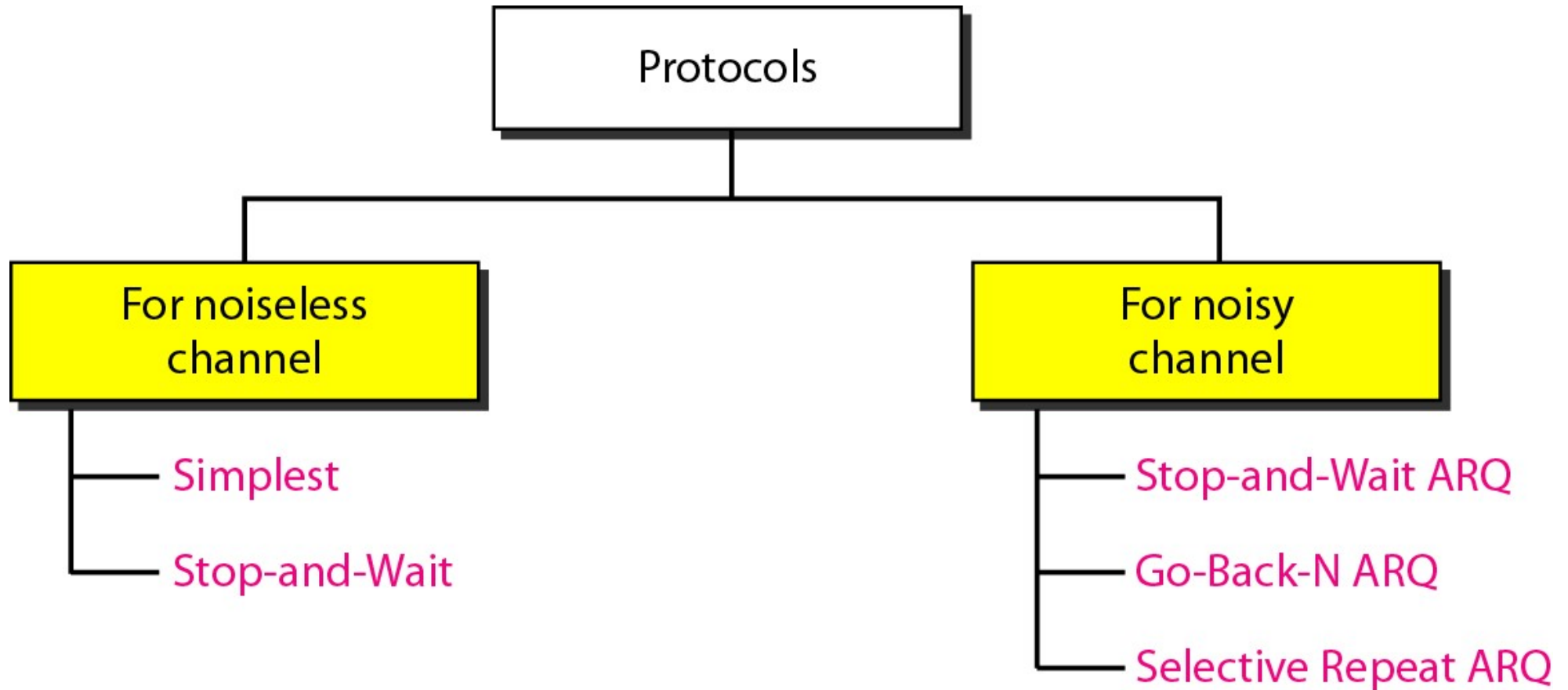
Note

Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.

Cont..

- Two approaches are commonly used.
 - In the first one, **feedback-based flow control**, the receiver sends back information to the sender giving it permission to send more data, or at least telling the sender how the receiver is doing.
 - In the second one, **rate-based flow control**, the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver.
- In this chapter we will study feedback-based flow control schemes, primarily because rate-based schemes are only seen as part of the transport layer

Flow control schemes



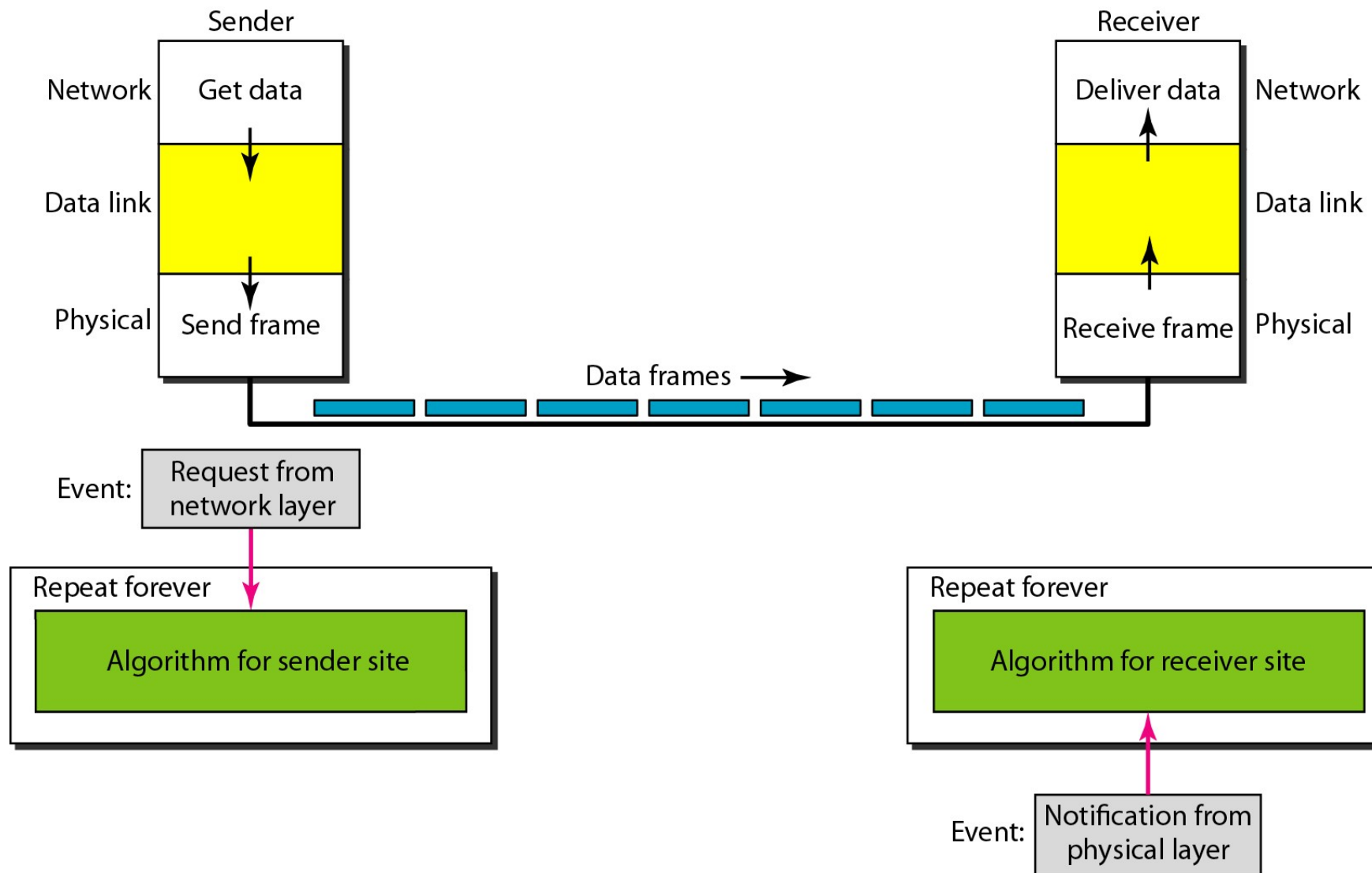
Noiseless channels

Let us first assume we have an ideal channel in which no frames are lost, duplicated, or corrupted.

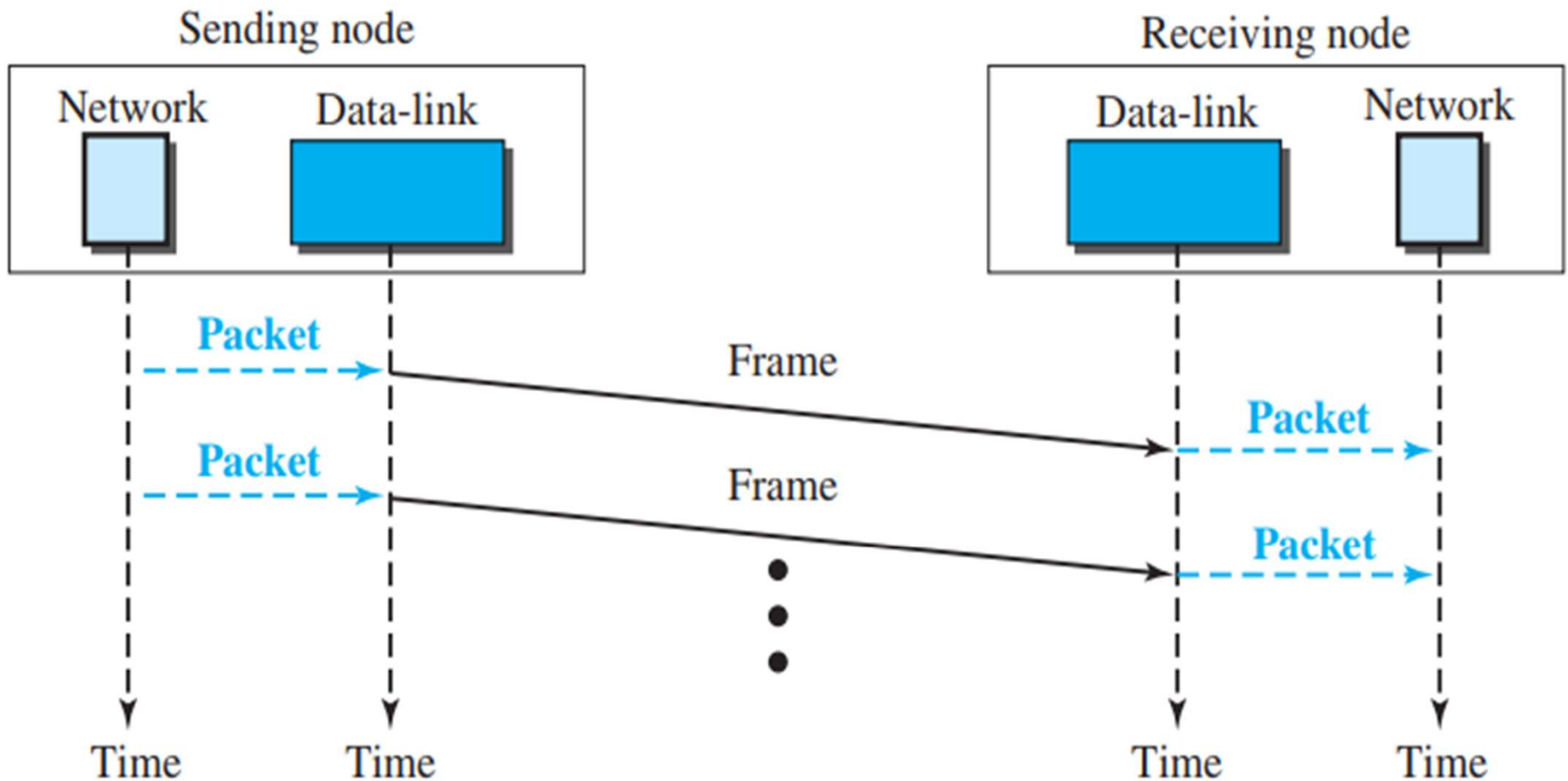
We introduce two protocols for this type of channel.

- **Simplest Protocol**
- **Stop-and-Wait Protocol**

The design of the simplest protocol with no flow or error control



Flow diagram



Design of Stop-and-Wait Protocol

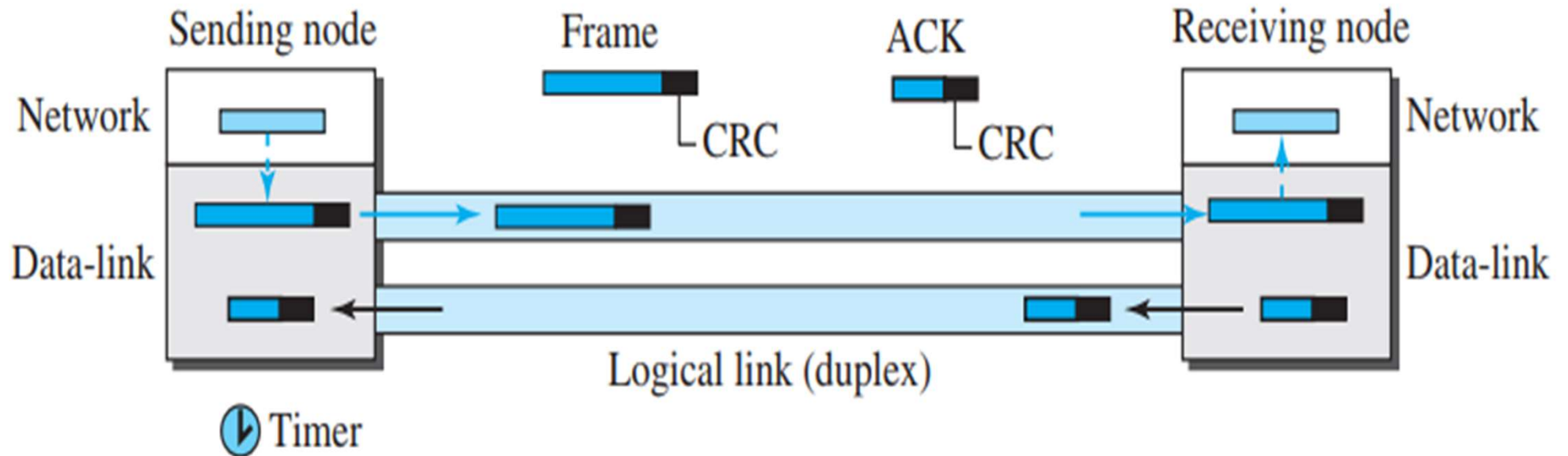
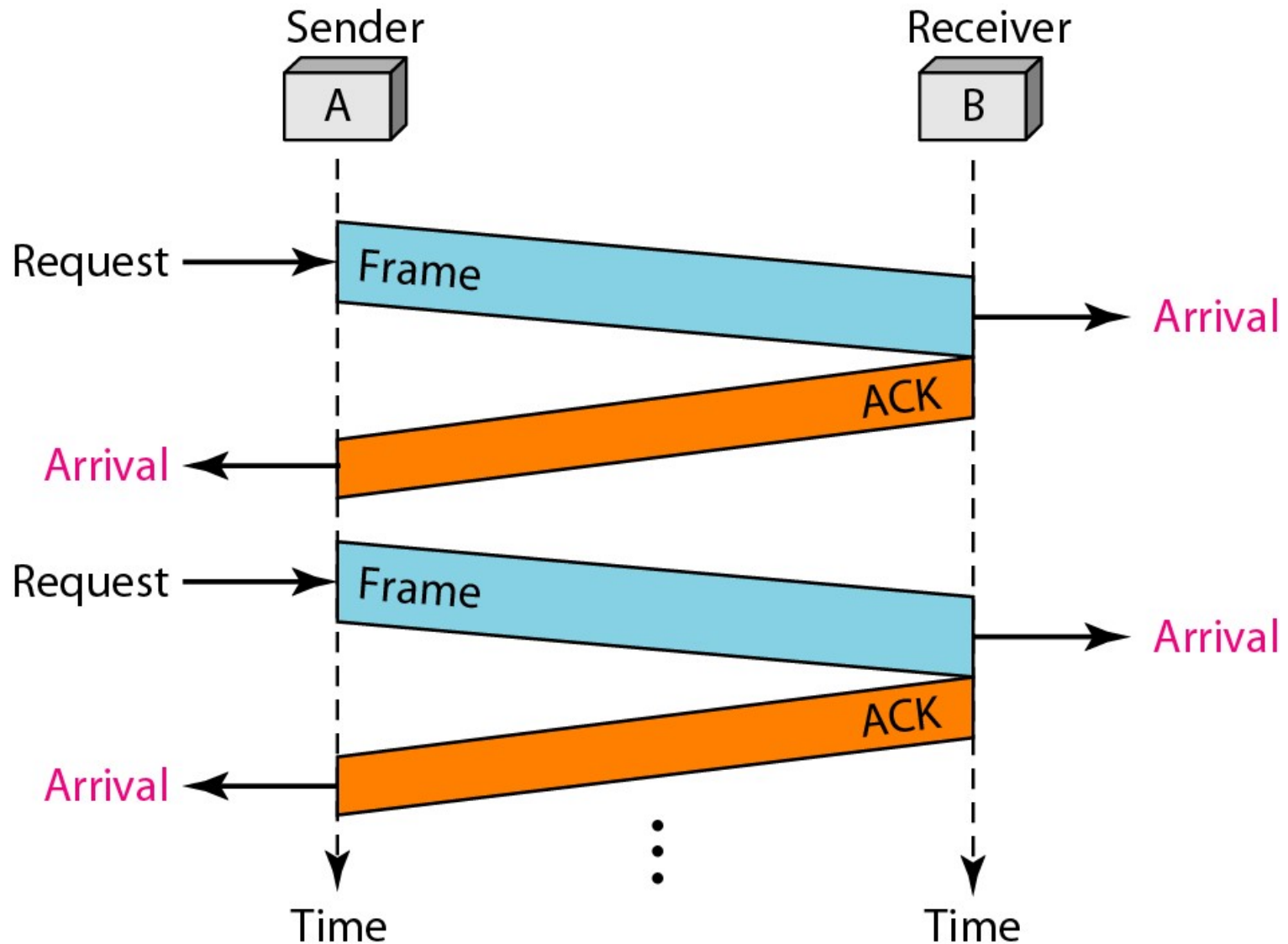


Figure 11.9 *Flow diagram for Example 11.2*



Noisy channels

- Although the Stop-and-Wait Protocol gives us an idea of how to add flow control to its predecessor, noiseless channels are nonexistent.
- We discuss three protocols in this section that use flow control.
 - ▣ Stop-and-Wait Automatic Repeat Request (ARQ)
 - ▣ Go-Back-N ARQ
 - ▣ Selective Repeat ARQ

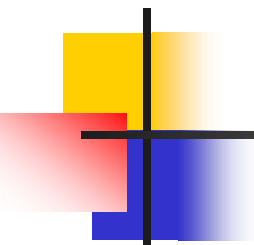
Automatic Repeat Request (ARQ) - any time an error is detected in an exchange, specified frames are retransmitted

Note

Error correction in Stop-and-Wait ARQ is done by keeping a copy of the sent frame and retransmitting of the frame when the timer expires.

Note

**In Stop-and-Wait ARQ, we use sequence numbers to number the frames.
The sequence numbers are based on modulo-2 arithmetic.**



Note

In Stop-and-Wait ARQ, the acknowledgment number always announces in modulo-2 arithmetic the sequence number of the next frame expected.

Design of the Stop-and-Wait ARQ Protocol

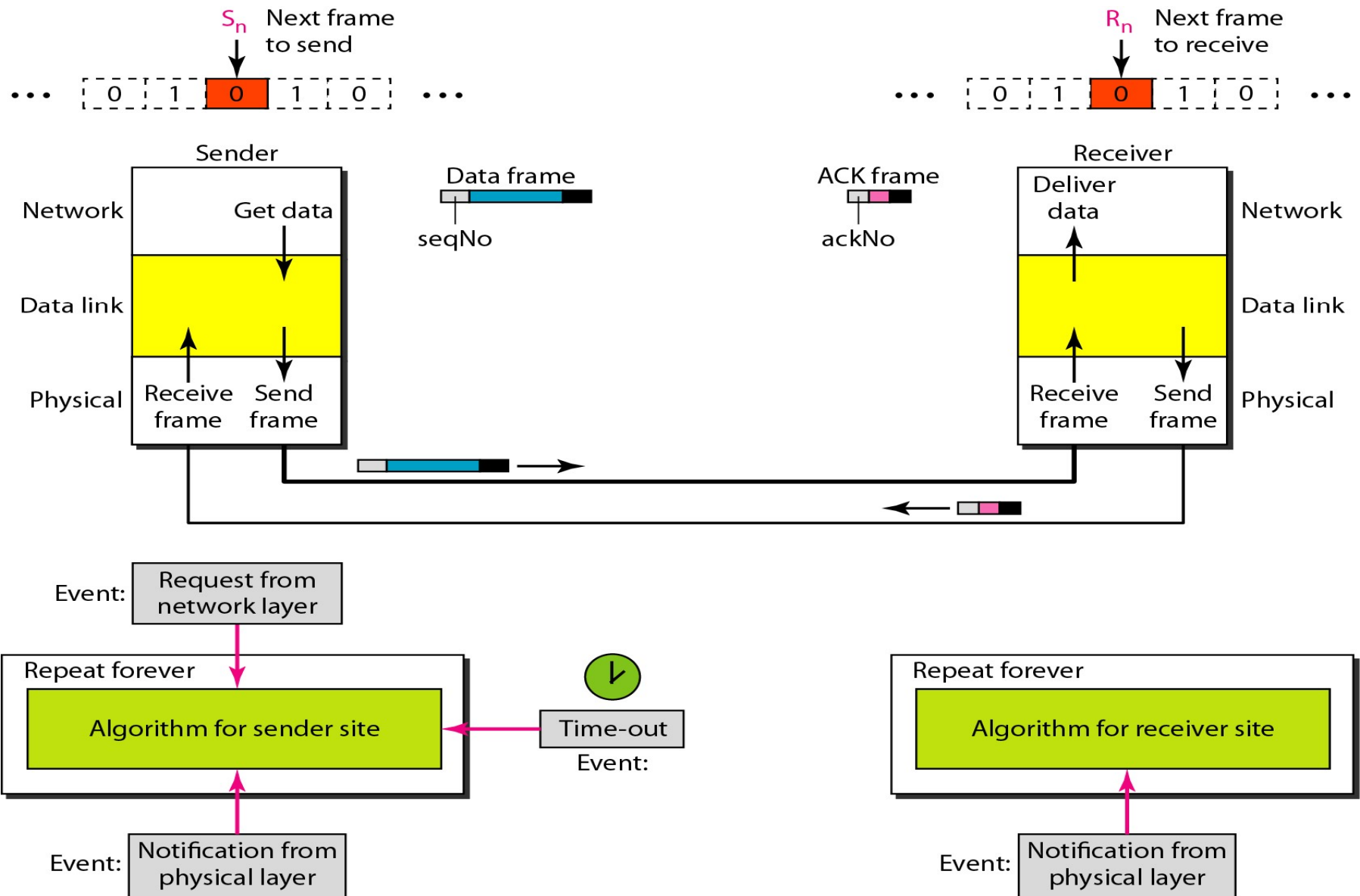
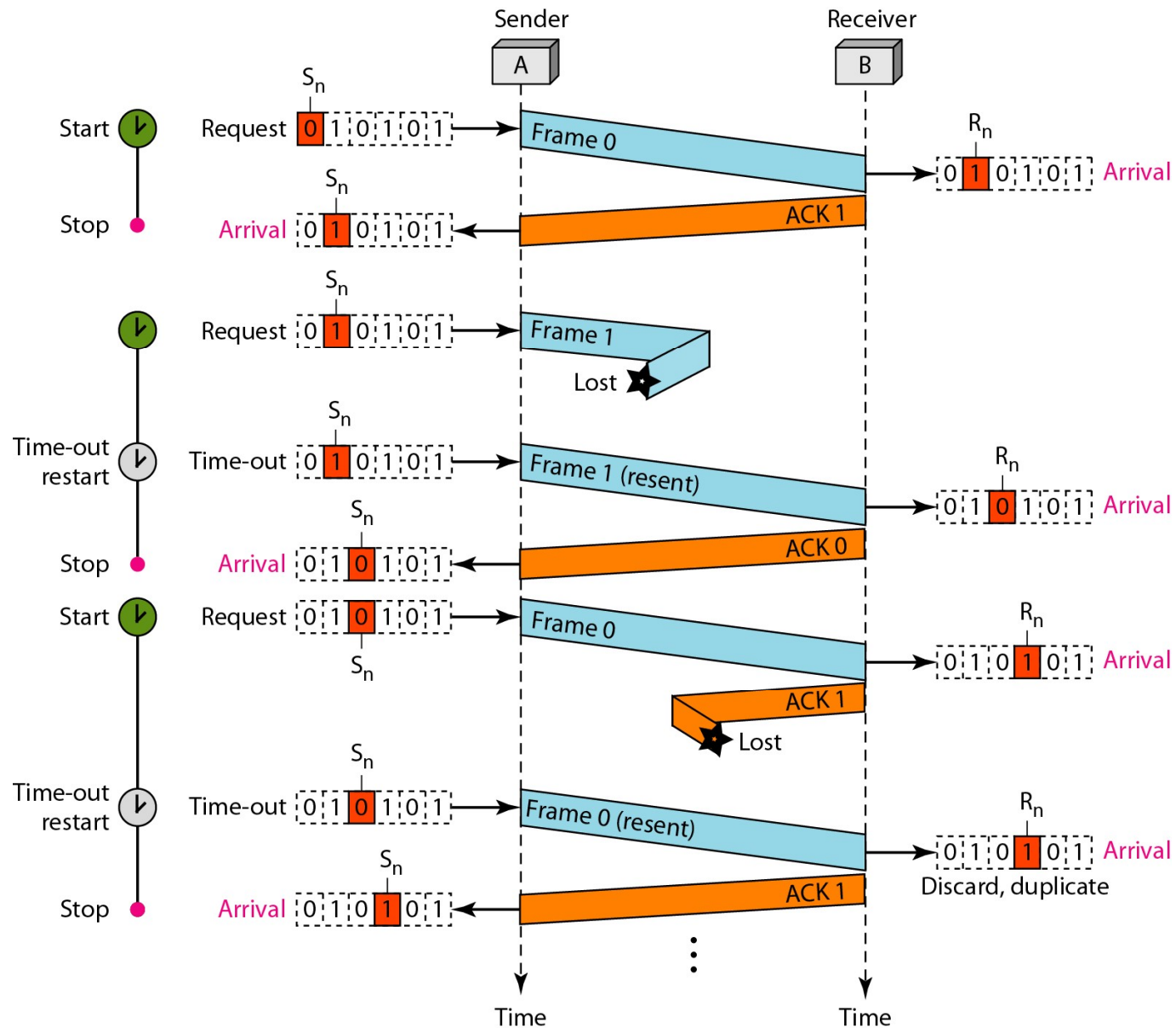


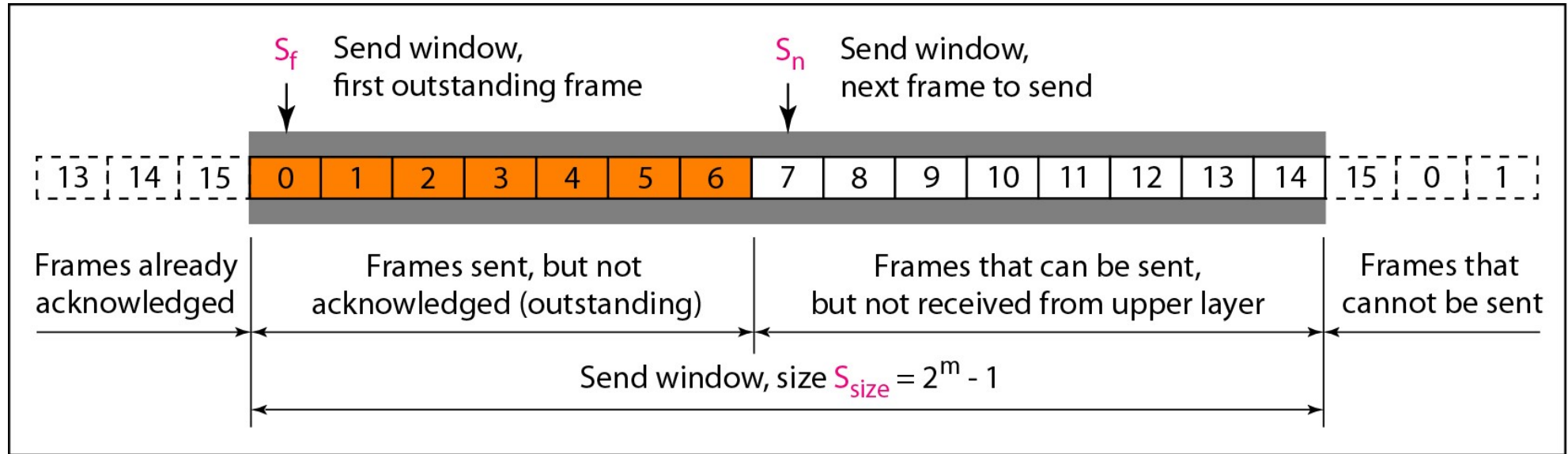
Figure 11.11 *Flow diagram for Example 11.3*



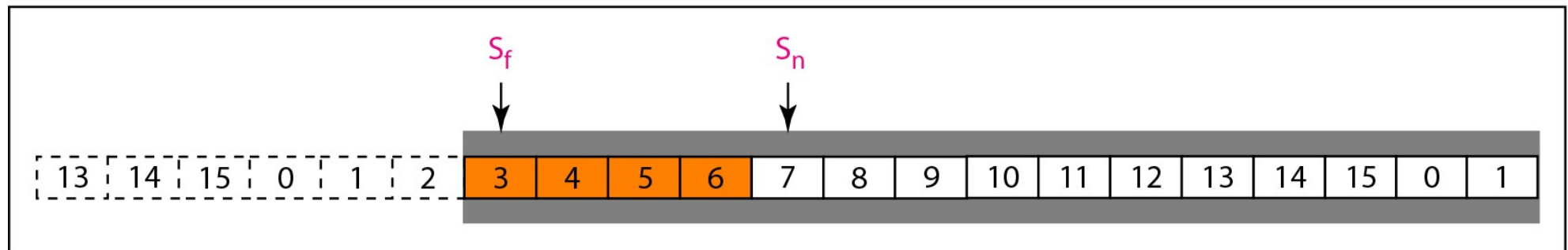
2. Go-Back-N ARQ

- In the Go-Back-N Protocol, the sequence numbers are **modulo 2^m** , where m is the size of the sequence number field in bits.
- **Sliding window** defines the range of sequence numbers that is the concern of the sender and receiver - sender and receiver deals with only a part of the range of sequence numbers
 - ▣ The send window is an abstract concept defining an imaginary box of size $2^m - 1$ with three variables: S_f , S_n , and S_{size} .
 - ▣ The send window can slide one or more slots when a valid acknowledgment arrives.

Figure 11.12 *Send window for Go-Back-N ARQ*



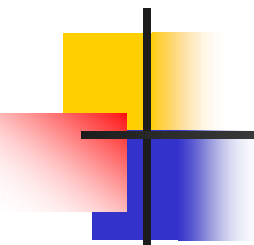
a. Send window before sliding



b. Send window after sliding

Note

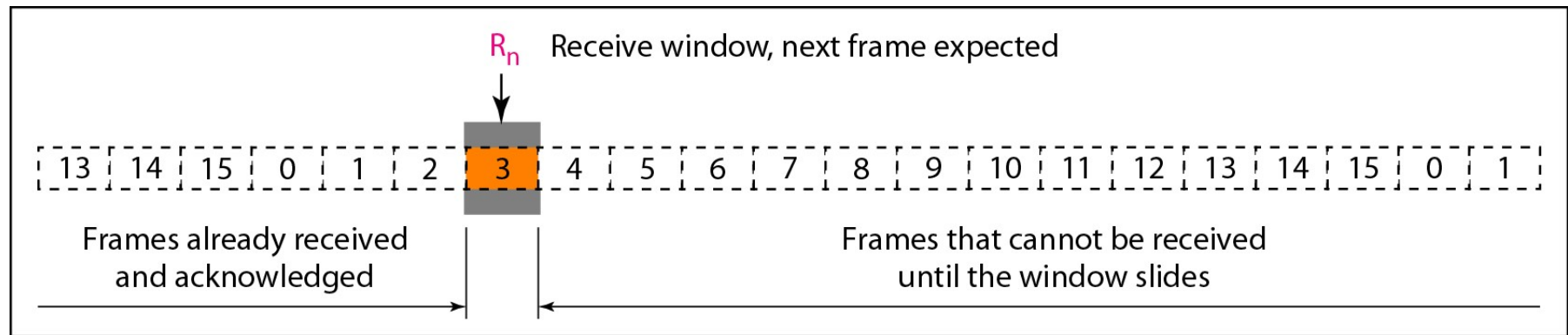
The send window is an abstract concept defining an imaginary box of size $2^m - 1$ with three variables: S_f , S_n , and S_{size} .



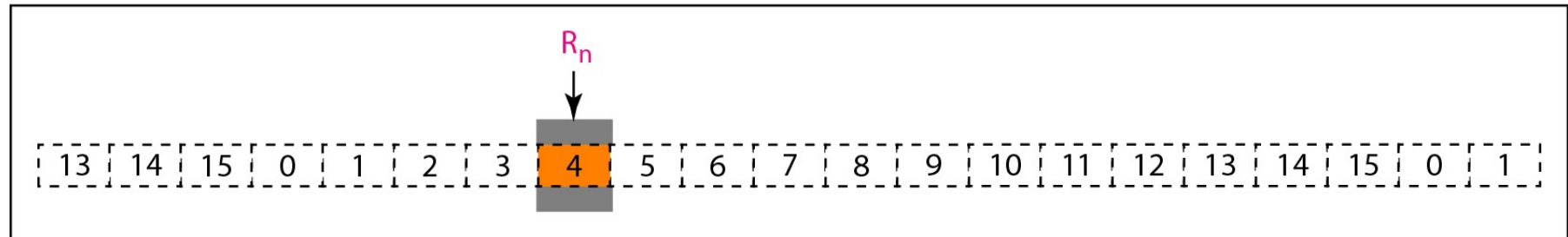
Note

The send window can slide one or more slots when a valid acknowledgment arrives.

Figure 11.13 *Receive window for Go-Back-N ARQ*



a. Receive window



b. Window after sliding

Figure 11.16 *Flow diagram for Example 11.6*

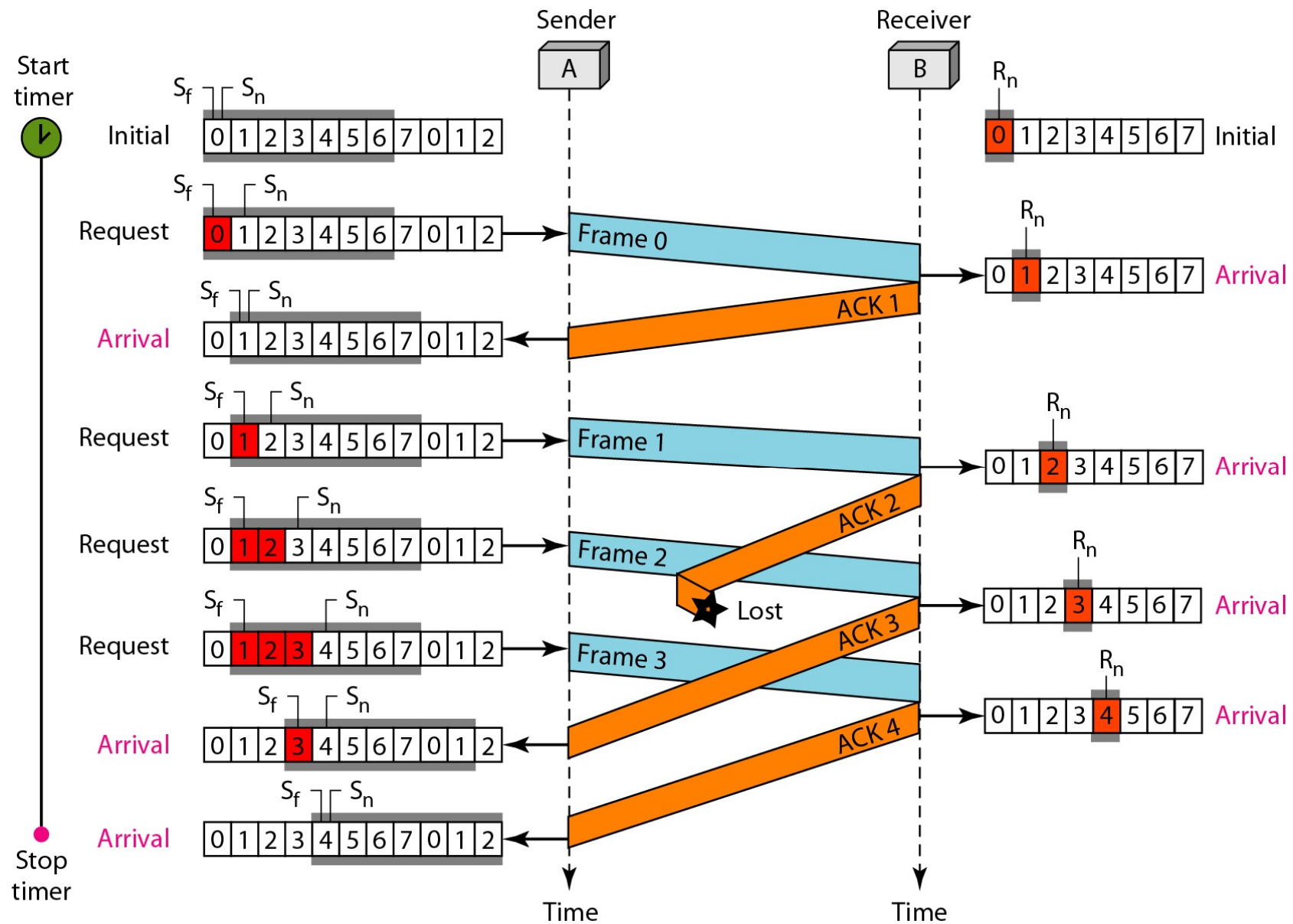
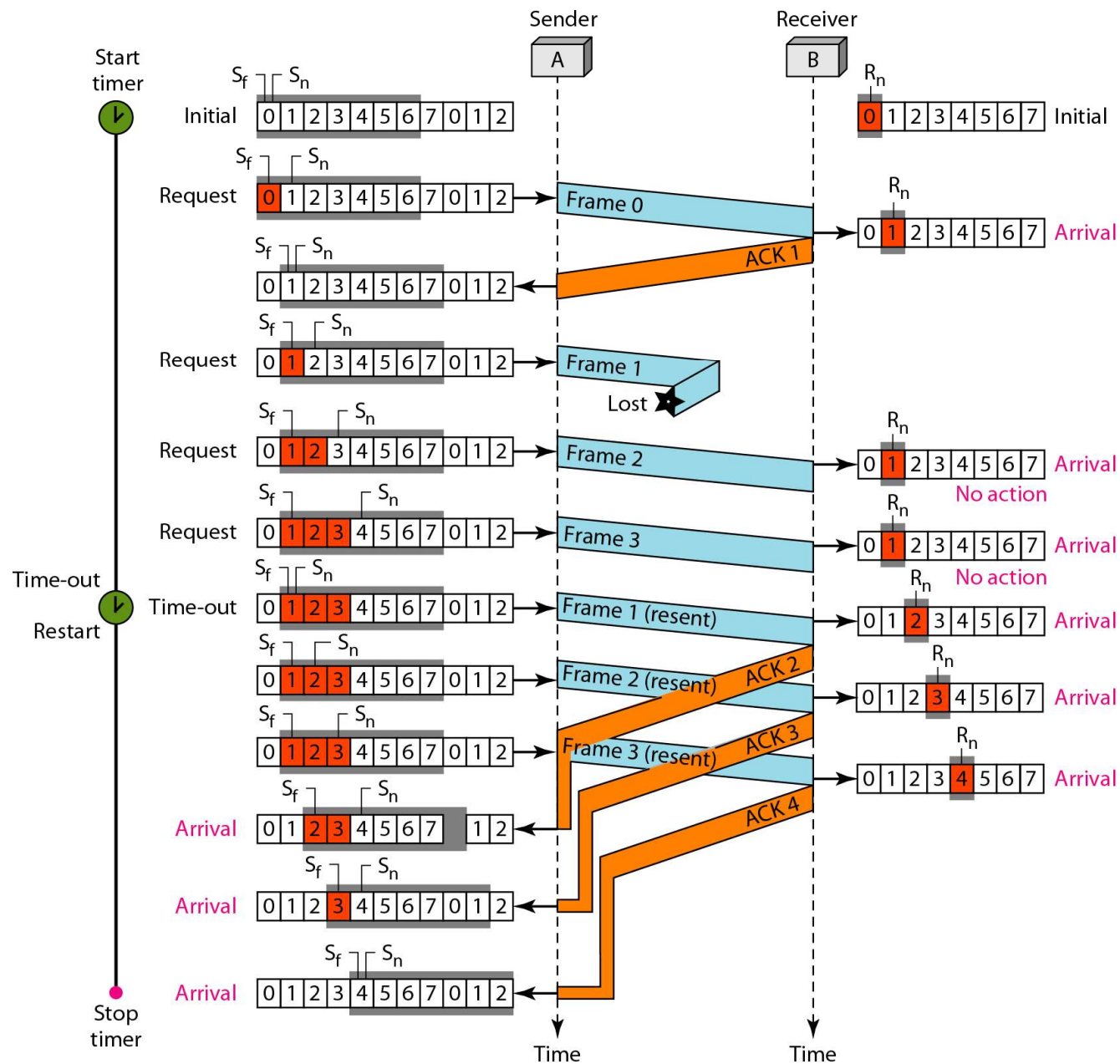


Figure 11.17 *Flow diagram for Example 11.7*



3. Selective Repeat ARQ

- *Go-Back-N* ARQ simplifies the process at the receiver site.
- The receiver keeps track of only one variable, and there is no need to buffer out-of-order frames; they are simply discarded.
- However, this protocol is very inefficient for a noisy link. In a noisy link a frame has a higher probability of damage, which means the resending of multiple frames.
- This resending uses up the bandwidth and slows down the transmission.

For noisy links, there is another mechanism that does not resend N frames when just one frame is damaged; only the damaged frame is resent.

- This mechanism is called Selective Repeat ARQ.

Figure 11.18 *Send window for Selective Repeat ARQ*

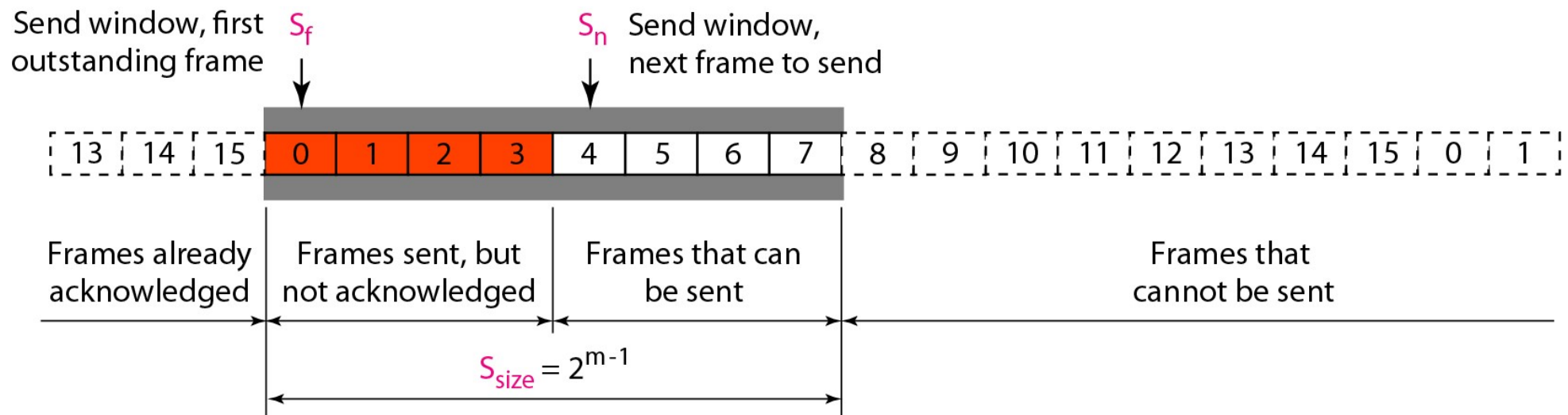


Figure 11.19 *Receive window for Selective Repeat ARQ*

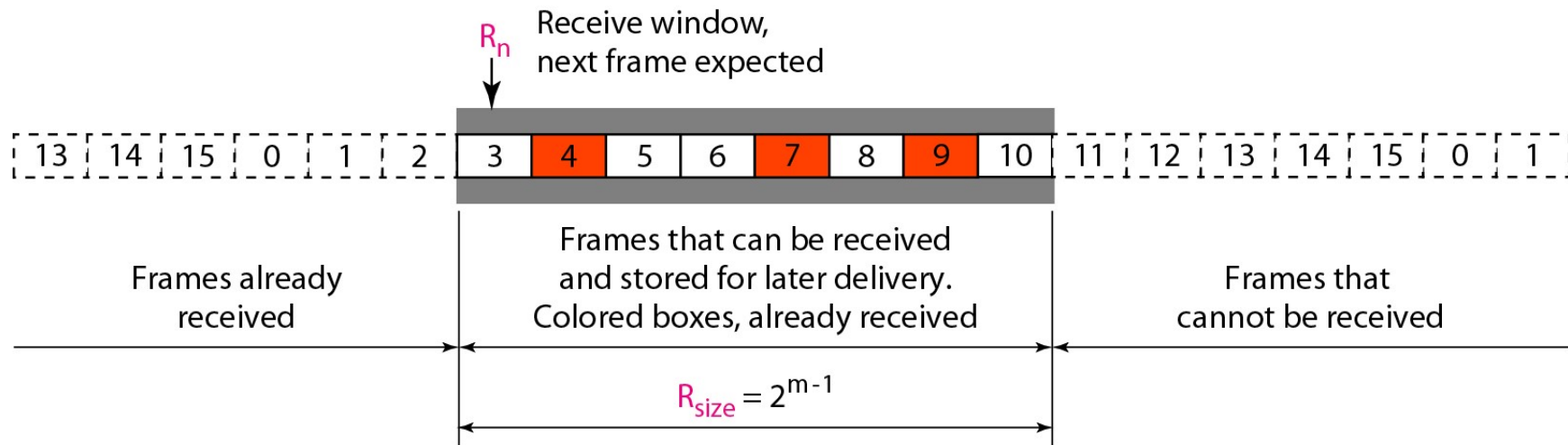


Figure 11.23 *Flow diagram for Example 11.8*

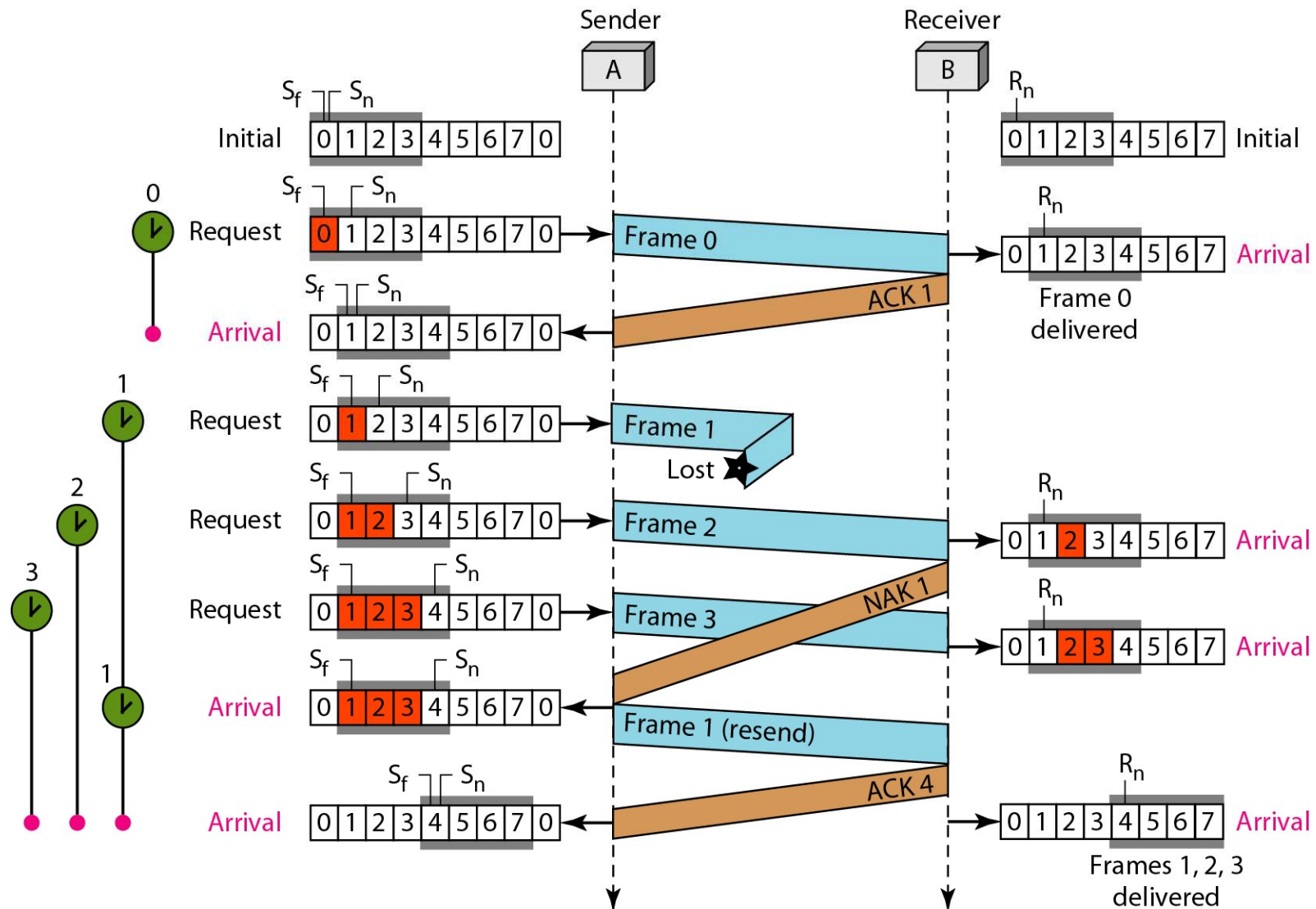


Figure 11.24 *Design of piggybacking in Go-Back-N ARQ*

